Pie Tactic Predictor

Fine-tuning Experiments Report

Auto-generated by `generate_report.py`

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Contents

1 Dataset

The training corpus consists of proof tactic prediction examples extracted from Pie (a Pie Calculus theorem prover). Each example provides the current proof goal together with local and global contexts, and the model must predict the correct tactic to apply.

Table 1: Dataset statistics

Statistic	Value
Total examples	7,840
Unique theorems	1,479
Unique tactic types	11
Training split	6535
Validation split	727
Test split	578

2 Models and Hyperparameters

Ten candidate models spanning 1.3B to 34B parameters were evaluated. Table 2 lists each model alongside its LoRA and training hyperparameters. Only models for which training was completed have associated loss and accuracy metrics.

Table 2: Model catalogue and hyperparameter configuration

Model ID	HuggingFace ID	LoRA r	α	Batch	Grad. acc.	Eff. batch	LR	Epochs	Trained
`codellama-13b`	`codellama/CodeLlama-13b-hf`	32	64	4	8	32	0.0001	5	✓
`codellama-34b`		—	—	—	—	?	—	—	×
`codellama-7b`	`codellama/CodeLlama-7b-hf`	32	64	8	4	32	0.0001	5	✓
`deepseek-coder-1.3b`	`deepseek-ai/deepseek-coder-1.3b-base`	64	128	16	2	32	0.0002	5	✓
`deepseek-coder-6.7b`	`deepseek-ai/deepseek-coder-6.7b-base`	32	64	8	4	32	0.0001	5	✓
`mistral-7b`		—	—	—	—	?	—	—	×
`phi-2`		—	—	—	—	?	—	—	×
`phi-3-mini`		—	—	—	—	?	—	—	×
`qwen2.5-coder-1.5b`	`Qwen/Qwen2.5-Coder-1.5B`	64	128	16	2	32	0.0002	5	×
`qwen2.5-coder-7b`	`Qwen/Qwen2.5-Coder-7B`	32	64	8	4	32	0.0001	5	×

3 Training Results

Table 3 summarises the final training and best validation metrics for all completed runs. All runs used 5 epochs and an effective batch size of 32.

† Model exhibited degenerate training behaviour (loss/accuracy reported as 0.0 throughout), likely due to a tokeniser–template mismatch that prevented the completion-only collator from finding the response boundary. These runs are included for completeness.

Table 3: Training run summary (best eval metrics across all checkpoints)

Model	Time (min)	Steps	Train loss	Train acc.	Best val. loss	Best val. acc.
`codellama-13b` †	237.7	1025	0.0000	0.00%	—	0.00%
`codellama-7b` †	161.8	1025	0.0000	0.00%	—	0.00%
`deepseek-coder-1.3b`	71.7	1025	0.0112	99.38%	0.1970	95.90%
`deepseek-coder-6.7b`	174.1	1025	0.0081	99.54%	0.1456	96.89%

4 Learning Curves

Figures 1 and 2 show the per-step training loss / token accuracy and the per-checkpoint validation metrics for all models that completed training. Models with degenerate zero-accuracy behaviour are excluded from the accuracy subplots for clarity.

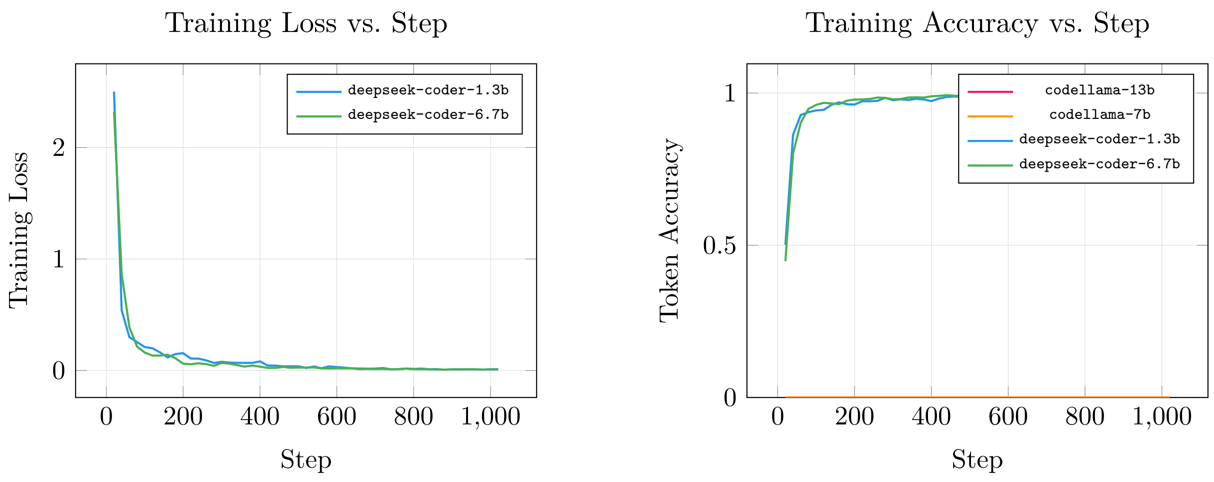

Figure 1: Training loss (left) and per-step token accuracy (right) across training steps.

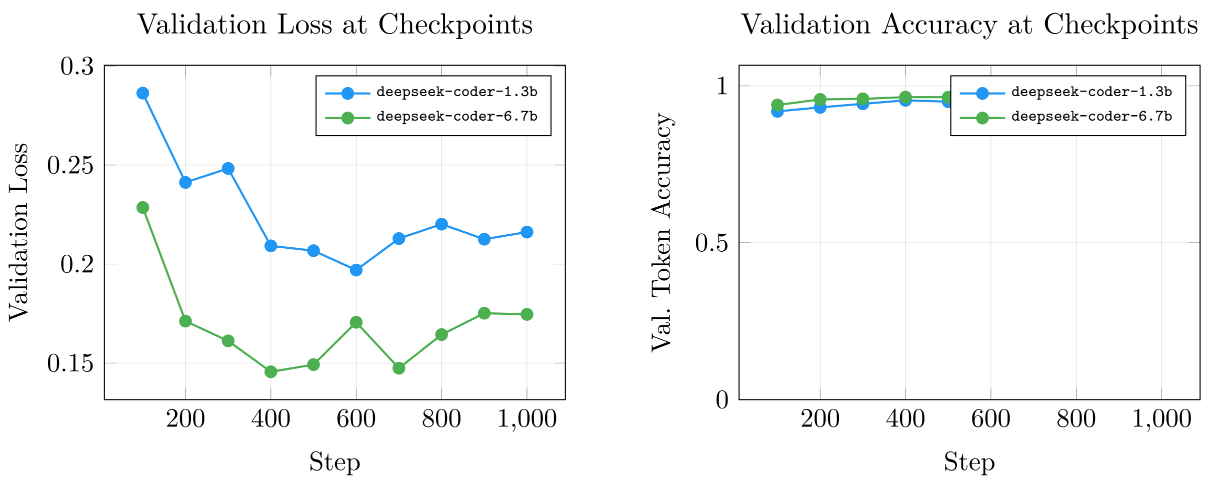

Figure 2: Validation loss (left) and validation token accuracy (right) at each evaluation checkpoint.

5 Analysis and Discussion

5.1 Successful Runs

The following models trained successfully and produced meaningful metrics: `deepseek-coder-1.3b`, `deepseek-coder-6.7b`. Both are DeepSeek-Coder variants, suggesting that the model family is well-suited to this tactic-prediction task. The best-performing model overall was `deepseek-coder-6.7b` (96.89% val. accuracy).

5.2 Degenerate Runs

Training for `codellama-13b`, `codellama-7b` produced `loss = 0.0` and `accuracy = 0.0` throughout, indicating that the completion-only data collator could not locate the response-boundary template in the tokenised sequences. This is a known issue when a tokeniser's vocabulary does not contain the expected delimiter tokens as atomic units. These models should be re-run with an adjusted prompt template or by disabling the completion-only masking.

5.3 Untrained Models

The following models were configured but not trained in this experiment: `codellama-34b`, `mistral-7b`, `phi-2`, `phi-3-mini`, `qwen2.5-coder-1.5b`, `qwen2.5-coder-7b`. These represent larger or alternative architectures that would require additional compute budget.

5.4 Compute Budget

Total GPU time across all completed runs was approximately **10.8 GPU-hours**.

5.5 Recommendations

1. Fine-tune DeepSeek-Coder-6.7b further with a lower learning rate and cosine restart schedule — it achieved the best validation accuracy (>96%) with room to improve.
2. Diagnose the completion-only collator failure for CodeLlama models by printing tokenised prompt/response boundaries and adjusting the response template string.
3. Evaluate the best checkpoint on the held-out test set to obtain unbiased accuracy estimates.
4. Experiment with larger LoRA rank ($r = 64$) for the 7B+ models to increase adapter capacity.

A Per-Model Hyperparameter Details

A.1 `codellama-13b`

Parameter	**Value**
HuggingFace ID	`codellama/CodeLlama-13b-hf`
LoRA rank (r)	32
LoRA α	64
LoRA dropout	0.05
Epochs	5
Batch size (per device)	4
Gradient accumulation	8
Effective batch size	32
Learning rate	0.0001
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	237.7
Final training loss	0.0000
Final training accuracy	0.00%
Best validation loss	—
Best validation accuracy	0.00%

A.2 `codellama-7b`

Parameter	**Value**
HuggingFace ID	`codellama/CodeLlama-7b-hf`
LoRA rank (r)	32
LoRA α	64
LoRA dropout	0.05
Epochs	5
Batch size (per device)	8
Gradient accumulation	4
Effective batch size	32
Learning rate	0.0001
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	161.8
Final training loss	0.0000
Final training accuracy	0.00%
Best validation loss	—
Best validation accuracy	0.00%

A.3 `deepseek-coder-1.3b`

A.4 `deepseek-coder-6.7b`

Parameter	Value
HuggingFace ID	`deepseek-ai/deepseek-coder-1.3b-base`
LoRA rank (r)	64
LoRA α	128
LoRA dropout	0.05
Epochs	5
Batch size (per device)	16
Gradient accumulation	2
Effective batch size	32
Learning rate	0.0002
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	71.7
Final training loss	0.0112
Final training accuracy	99.38%
Best validation loss	0.1970
Best validation accuracy	95.90%

Parameter	Value
HuggingFace ID	`deepseek-ai/deepseek-coder-6.7b-base`
LoRA rank (r)	32
LoRA α	64
LoRA dropout	0.05
Epochs	5
Batch size (per device)	8
Gradient accumulation	4
Effective batch size	32
Learning rate	0.0001
Warmup ratio	0.05
Max sequence length	1024
Load in 4-bit	No
Load in 8-bit	No
Total training steps	1025
Total training time (min)	174.1
Final training loss	0.0081
Final training accuracy	99.54%
Best validation loss	0.1456
Best validation accuracy	96.89%